Collective Intelligence in Architectural Programming

Multi-Agent Reinforcement Learning for Optimization in Mega Transportation Projects

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Abstract. Mega transportation projects (MTPs), as critical urban nodes, rely on architectural programming to define their objectives and pathways. Existing approaches mainly depend on static deduction mechanisms, which are inadequate for addressing the strategic interactions and dynamic demands among stakeholders. This study proposes a multi-agent reinforcement learning-based framework for architectural programming optimization. First, a multi-agent proximal policy optimization model is developed to simulate the preferences and behaviours of diverse agents. Second, a counterfactual regret-minimization mechanism and a hierarchical reinforcement learning architecture are integrated to enhance the efficiency of conflict mediation and facilitate multi-objective decomposition. This study is validated using Hangzhou West Railway Station in China as a case study. Results demonstrate that the framework significantly outperforms baseline models across eight architectural programming dimensions: function, form, economy, time, humanity, environment, culture, and safety. This study not only provides an intelligent approach to architectural programming in MTPs but also establishes a more resilient paradigm for addressing uncertainty in the architecture, engineering, and construction industry.

Keywords. Mega transportation projects, architectural programming, multi-agent reinforcement learning, proximal policy optimization

1. Introduction

Serving as critical infrastructure, mega transportation projects (MTPs) not only enhance travel efficiency and optimize traffic flow but also exert profound impacts on spatial patterns, economic growth, and social sustainability (Li et al., 2024). With accelerating urbanization, cities face escalating challenges such as exacerbating congestion, inequitable resource distribution, and environmental pollution. Therefore, MTPs are considered a crucial means to bolster urban resilience, promote regional synergistic development, and drive industrial upgrading (Javed et al., 2022).

Architectural programming, as the starting point of the MTP lifecycle, has a

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profound influence on subsequent design and construction phases (Jin, 2024). Its core task is to integrate the needs of different stakeholders, ultimately achieving rational allocation and forward-looking guidance (Hershberger, 2015). However, architectural programming for MTPs confronts challenges, including the conflict of stakeholder objectives, the uncertainty of the programming environment, and the complexity of multi-objective optimization.

Collective intelligence offers a new pathway for the optimization of architectural programming. Represented by multi-agent reinforcement learning (MARL), it can not only model the interaction process among multiple agents but also achieve game-theoretic outcomes in a distributed framework to arrive at a globally optimal solution for the entire system (He et al., 2022). In the AEC domain, some studies have attempted to utilize MARL for tasks like optimizing urban functional zoning, generating architectural design parameters, and scheduling construction processes (Jin, 2025; Yao et al., 2024).

Building upon this foundation, this study proposes a multi-agent reinforcement learning (MARL) framework for architectural programming in MTPs. Compared with existing studies, the contributions are threefold: (1) architectural programming is formally conceptualized as a multi-agent game, enabling explicit modelling of information heterogeneity and behavioural preference divergence among multiple stakeholders; (2) multi-agent proximal policy optimization (MAPPO) is integrated with counterfactual regret minimization (CFR) to address nonlinear strategic conflicts and local-optimum traps in multi-agent decision-making; (3) a three-level decision-making architecture based on hierarchical reinforcement learning (HRL) is constructed to achieve dynamic coupling and decomposed optimization across strategic objectives, collaborative policies, and execution-level actions.

2. Literature Review

MTPs must not only meet fundamental objectives but also integrate multi-dimensional factors, including social, cultural, and humanistic values (Milovanovic et al., 2023). Programming approaches based on big data enable quantitative demand forecasting and scheme evaluation (Zhuang et al., 2023). Building information modelling (BIM) and city information modelling (CIM) provide a unified digital framework for this purpose. By integrating traffic flow, land use, and environmental quality, they facilitate multi-scenario assessments of programming schemes (Gu et al., 2023). Multi-objective optimization methods offer novel solutions for addressing the complex trade-offs inherent in megaprojects' programming. Such as NSGA-II and MOEA/D, search for Pareto optimal solution sets by simulating natural selection processes, thereby achieving a balance among different objectives (Casotti and Zio, 2025). Collaborative programming based on intelligent agent systems has emerged as a significant research direction in recent years. This approach treats different stakeholders as agents with independent objectives and strategic boundaries. Simulating their interaction processes reveals the system's collaborative patterns and conflict mechanisms (Song et al., 2025).

Reinforcement learning (RL), unlike supervised learning that relies on labelled data, develops a policy evolution mechanism through trial-and-error learning in dynamic environments (Shakya et al., 2023). This characteristic has endowed it with unique advantages in adaptive optimization and behavioural coordination for complex systems (Zhang et al., 2023). Wu et al. (2025) utilized RL to control building heating systems, achieving synergistic optimization of energy efficiency and occupant comfort. However, single-agent optimization often fails to capture the multi-agent game-theoretic relationships. To address this, researchers have proposed MARL to model the collaboration of multiple agents within a shared environment. Li (2025) constructed a hierarchical MARL framework that mapped the design phase, cost control, and environmental performance to different levels of policy networks, thereby achieving the maximization of long-term benefits. To balance individual autonomy and system-wide coordination, MARL has further evolved to incorporate the centralized training with decentralized execution (CTDE) architecture. This paradigm allows agents to make independent decisions based on local information while achieving global coordination through a centralized value function (Han et al., 2024).

Despite the demonstrated potential of RL in collaborative optimization, their application in MTPs still faces challenges: (1) Current optimization frameworks are often based on rule-driven simulation systems, which struggle to effectively balance individual gains with global objectives in complex environments; (2) When dealing with multi-stage objective coupling, sparse long-term rewards, and non-linear policy conflicts, these methods often suffer from slow convergence and poor stability.

3. Methodology

3.1. PROBLEM FORMULATION

According to the value-based theory of architectural programming proposed by Hershberger (2015), the key programming dimensions of MTPs can be categorized into eight interrelated aspects: function, form, economy, time, humanity, environment, culture, and safety. Transportation functionality serves as the core constraint, requiring that the project's various components satisfy functional demands for traffic circulation and collection, transfer connectivity, commercial services, and operational management. Furthermore, the scheme must be feasible in terms of massing control, spatial organization, and structural logic, and it must comply with relevant safety regulations, including egress distances, safety exit layouts, fire compartmentation, and crowd load control. In addition, the primary objectives also pertain to humanistic perception and environmental impact, focusing not only on user experience and comfort but also on enhancing sustainable performance, such as natural ventilation and daylighting. Another critical objective is economic viability, achieved through efficient land utilization and mixed-use functional design. The cultural significance of the architecture also represents a vital objective, emphasizing integration with the urban context and local landscape to embody long-term value. Based on these considerations, given a set of dimensions $D = \{d1, d2, \dots, dn\}$ and a state attribute S, the programming problem can be formally expressed as follows:

$$\max_{D} f(hum(D,S) + env(D,S) + eco(D,S) + cul(D,S) | w_1, w_2, w_3, w_4) \quad (1)$$

$$function(d_i, S) = True, \forall i \in \{1,2,\dots,n\} \quad (2)$$

$$s.t.\ d_i \cap S = \emptyset, \forall i \in \{1,2,\dots,n\} \quad (3)$$

$$safety(d_i|S) \leq S_{max}, \forall i \in \{1,2,\dots,n\} \quad (4)$$

$$time(d_i|S) \leq T_{limit}, \forall i \in \{1,2,\dots,n\} \quad (5)$$

Where Eq. (1) defines the overall objective function, including humanity, environment, economy, and culture; Eq. (2) represents the functional constraints of MTPs; Eq. (3) specifies the feasibility constraints related to architectural form and spatial organization; Eq. (4) enforces safety regulations; and Eq. (5) constrains the temporal controllability of the programming scheme.

3.2. MULTI-AGENT MODELLING

The stakeholder set of MTPs is $A = \{a_1, a_2, \ldots, a_N\}$, where N is the number of agents participating in the programming process. Each agent a_i represents a specific type of stakeholder, including governments (a_g), investors (a_i), designers (a_d), constructors (a_c), and users (a_u). Based on the responsibilities and strategic scopes of each agent, this study constructs their independent action spaces (Table 1).

Table 1. Action Spaces and State Impacts

Agent	Action	Affected Dimensions
Government	Land Allocation Coefficient Adj.	$\Delta func, \Delta eco, -\Delta env$
	Planning Intensity / FAR Control	$\Delta form, \Delta safe$
	Environmental Policy Stringency	$\Delta env, \Delta cul$
Investor	Investment Allocation Ratio	$\Delta eco, -\Delta time$
	Risk Tolerance	$\Delta eco, -\Delta safe$
Designer	Functional Zoning Adjustment	$\Delta func, \Delta form, \Delta hum$
	Green Design Intensity	$\Delta env, -\Delta eco$
Constructor	Construction Pace Adjustment	$\Delta time, -\Delta safe, \Delta eco$
	Resource Dispatching Intensity	$\Delta eco, -\Delta env$
User	Traffic Accessibility Demand	$\Delta func, \Delta hum, \Delta cul$
	Comfort Preference	$\Delta hum, \Delta env$

The reward function is the feedback signal provided by the MTP programming environment to the agents, guiding them to learn the optimal collaborative policy. To balance individual objectives with overall system performance, this study constructs a multi-level reward structure: individual reward, global reward, and constraint penalty:

$$r_i^{local}(t) = f_i(\Delta E_i, \Delta C_i, \Delta T_i, \Delta S_i) \quad (6)$$

$$R_t^{global} = w_1 Q_{env} + w_2 Q_{eco} + w_3 Q_{hum} + w_4 Q_{cul} \quad (7)$$

$$P_t = -\lambda_1 max(0, S_t - S_{max}) - \lambda_2 max(0, T_t - T_{limit}) \quad (8)$$

ΔE_i denotes the improvement in energy efficiency, ΔC_i is the cost savings rate, ΔT_i is the reduction in construction time, and ΔS_i is the change in social satisfaction. Q_{env} 、Q_{eco}、Q_{hum}, and Q_{cul} respectively correspond to the overall performance of the environment, economy, humanity, and culture. λ_1 and λ_2 are penalty coefficients used to suppress the emergence of non-compliant strategies.

3.3. POLICY OPTIMIZATION

After modelling the agents, this study utilizes the PPO to facilitate dynamic game-theoretic learning within the architectural programming scenario. During the centralized training phase, agents have access to the global state S_t and the joint action $A_t = \{a_1^t, a_2^t, \ldots, a_N^t\}$, and a centralized value network is used to construct a global value function $Q(S_t, A_t)$. In the decentralized execution phase, each agent makes decisions independently based on its own local observation o_i^t and outputs an action distribution $\pi_{\theta_i}(a_i^t|o_i^t)$ using its policy network. During this process, policy updates are constrained by a clipped surrogate objective function, which limits the magnitude of change between the old and new policies to prevent excessive updates:

$$L(\theta) = \mathbb{E}_t[min(r_t(\theta)A_t, clip(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)] \tag{9}$$

where $r_t(\theta) = \pi_\theta(a^t|s^t)/\pi_{\theta_{old}}(a^t|s^t)$ is the ratio of the new policy to the old policy, A_t is the advantage function, and ϵ is the clipping hyperparameter. Although PPO can stably optimize individual policies, its update direction may still converge to a local optimum. To address this, this study incorporates the CFR mechanism to guide the direction of policy evolution from a game-theoretic equilibrium perspective. CFR quantifies and corrects behavioural deviations by calculating the "regret," the opportunity loss between the action actually taken and the best potential action in the current state. The regret for agent i at time t is defined as:

$$R_i^t = \max_{a_i'} Q_i(s_i^t, a_i') - Q_i(s_i^t, a_i^t) \tag{10}$$

A large value of R_i^t indicates that the agent's action has deviated significantly from the optimal behavior in that state. The system then increases the sampling probability of high-value actions in the next iteration, thereby progressively minimizing the long-term cumulative regret. This process functions as a policy correction mechanism based on counterfactual reasoning, enabling agents to gradually converge toward an approximate Nash equilibrium through adaptive adjustments.

3.4. HIERARCHICAL ARCHITECTURE

To accurately simulate the hierarchical decision-making structure of stakeholders in MTPs, this study employs HRL to abstract the architectural programming into the strategic layer, the collaborative layer, and the execution layer. At the strategic layer, the government and investors act as the dominant agents $a^{(H)}$, whose task is to establish objectives and constraint boundaries. The decision variables are denoted as $g_t^{(H)} \in \mathcal{G}$, which are optimized by maximizing the value function $V^{(H)}(S_t)$ to achieve long-term sustainability. Formally, the constraint imposed by the state-action pair can be expressed as:

$$C^{(H)}\big(a^{(M)}, S_t\big) = A^{(M)}a^{(M)} - \Omega^{(H)}\left(g_t^{(H)}\right) \leq 0 \tag{11}$$

where $A^{(M)}$ is the coefficient matrix for the collaborative layer. If any constraint is not satisfied, the collaborative layer action is deemed infeasible. The agents at the collaborative layer, $a^{(M)}$, correspond to designers and users. Their primary task is to decompose strategic layer objectives into operational policies. The state $S_t^{(M)} \subseteq S_t$ and action space $\mathcal{A}^{(M)}$ is defined by the constraints $\Omega^{(H)}\left(g_t^{(H)}\right)$ mapped from the strategic layer:

$$M\left(a^{(M)} \mid S_t, g_t^{(H)}\right) = M_{feas}\big(a^{(M)} \mid S_t\big) \cdot \mathbb{I}\left\{\exp\left[-\lambda D\left(a^{(M)}, g_t^{(H)}\right)\right] \geq \tau\right\} \tag{12}$$

where $D(\cdot)$ is the deviation between the action and the goal, λ is a sensitivity coefficient, τ is the masking threshold, and $\mathbb{I}\{\cdot\}$ is the indicator function. The execution layer agents, $a^{(L)}$, represent constructors and operators, their action space is typically continuous. Specifically, this layer's policy outputs the action distribution parameters $\mu_\theta, \sigma_\theta$, and actions are sampled within the feasible domain set by the execution layer:

$$a_{t+1}^{(L)} = \text{Proj}_{\Omega^{(H)}(a^{(M)})}\left(\mu_\theta\left(s_t^{(L)}\right) + \sigma_\theta \xi_t\right) \tag{13}$$

where ξ_t is a Gaussian noise term and the projection operator $\mathrm{Proj}_{\Omega^{(H)}(a^{(M)})}(\cdot)$ ensures that the action falls within the constraint domain.

4. Empirical Study

This study selects the Hangzhou West Railway Station Hub Complex as an empirical case to validate the effectiveness of the proposed framework (Figure 1).

Figure 1. Reasons for the selection of the empirical case

4.1. DATA COLLECTION AND SETTINGS

This study adopts a hybrid approach combining text mining and questionnaire surveys to quantify the key indicators and model parameters. First, a total of 58 textual documents related to the project, published between 2018 and 2022, were collected. Based on these materials, the relative importance scores of each programming dimension were computed using TF–IDF weighting. Second, a five-point Likert scale was employed to quantitatively assess the strategic preferences and reward weights of different stakeholder groups. A total of 96 valid questionnaires were obtained and used for statistical analysis. The final configurations are summarized in Table 2.

Table 2. Settings of state space, action space, and rewards

State space		Action space		Rewards	
S_{func}	[0.2,0.9]	Land allocation adjustment	[-0.2,0.2]	ΔE	0.26
S_{form}	[0.3,0.8]	Planning intensity / FAR control	[0.0,1.0]	ΔC	0.23
S_{eco}	[0.3,0.8]	Environmental policy stringency	[0.0,1.0]	ΔT	0.19
S_{time}	[0.0,0.3]	Investment allocation	[-0.3,0.3]	ΔS	0.32
S_{hum}	[0.2,0.9]	Risk tolerance	[0.0,1.0]	Q_{env}	0.25
S_{safe}	[0.7,1.0]	Functional zoning adjustment	[-0.2,0.2]	Q_{eco}	0.30
S_{env}	[0.4,0.9]	Green design intensity	[0.0,1.0]	Q_{hum}	0.25
S_{cul}	[0.2,0.8]	Construction pace adjustment	[-0.5,0.5]	Q_{cul}	0.20
		Resource scheduling	[-0.3,0.3]	λ_1	0.40
		Traffic accessibility demand	[0.0,1.0]	λ_2	0.30

Comfort preference	[0.0,1.0]

All experiments were conducted with the same configuration: AMD R7-9700X, Nvidia RTX 5090, and 32 GB RAM. The software consisted of Python 3.12, PyTorch 2.9, and CUDA 12.8. The hyperparameter settings are detailed in Table 3.

Table 3. Hyperparameter Settings

Hyperparameter	Value
Learning rate (Actor / Critic)	1e-4 / 1e-3
Discount factor (γ)	0.95
PPO clip range (ε)	0.2
Advantage estimation method	GAE ($\lambda = 0.95$)
CFR learning rate	Adaptive, initial value 0.1
Batch size	8192
Initial exploration rate	0.9
Final exploration rate	0.01
Exploration rate decay	0.0002

Five key metrics were recorded throughout the training process: global cumulative rewards, individual agent rewards, value loss, policy loss, and environmental state metrics. The experimental data were smoothed using a moving average to mitigate the impact of noise on trend analysis. The overall workflow is illustrated in Figure 2.

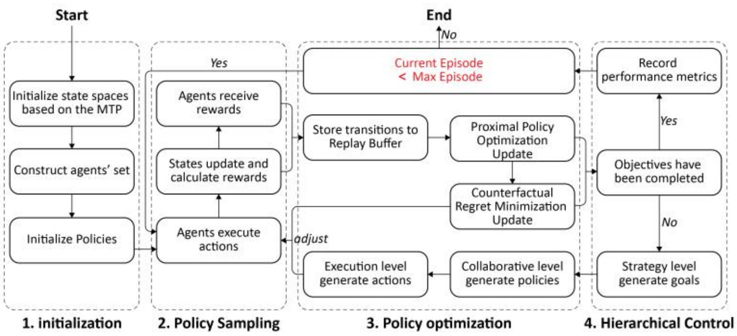

Figure 2. The workflow of the proposed framework.

4.2. CONVERGENCE AND OPTIMIZATION ANALYSIS

Figure 3 illustrates the trend of the global reward and the individual rewards. Overall, the global reward curve demonstrates rapid initial growth, followed by a period of oscillation before reaching a stable convergence. Individually, the government and investor agents, having clear objectives and well-defined constraints, exhibit reward curves that rise steadily and converge early. In contrast, the designer and constructor agents show fluctuations in the mid-training phase, reflecting the need to adjust their policies to manage trade-offs among multiple objectives. The user agent's reward is the most volatile, primarily due to the indirect influence of decisions made by other agents.

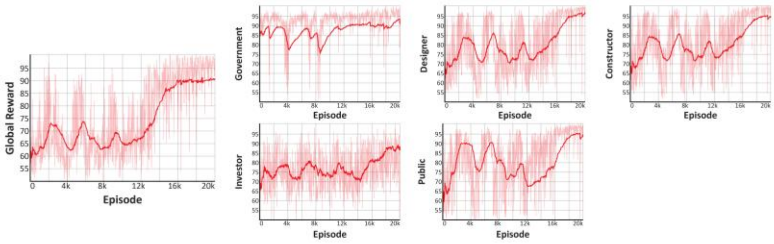

Figure 3. The global reward and individual rewards.

Table 4 presents the final state results, with eight core dimensions of architectural programming significantly improved compared to the initial state. The functional score indicates that the integrated relationship among transportation, commerce, and public space has been strengthened. The scores of the humanity and culture indicate an enhanced responsiveness of the project to public experience and regional culture. Furthermore, the state achieved synergistic growth in the economic and environmental dimensions, while the time and safety dimensions were maintained in a steady state, thereby validating the multi-objective optimization capabilities of the proposed framework.

Table 4. Collaborative Optimization Results and the Influence of Agent Behaviours

State value change		Influencing agent(s) and actions		
S_{func}	0.48→0.81	Government	Land allocation adjustment	+0.08
		Designer	Functional zoning adjustment	+0.15
S_{form}	0.52→0.77	Designer	Functional zoning adjustment	+0.25
S_{eco}	0.42→0.71	Investor	Investment allocation	+0.18
		Constructor	Construction pace adjustment	+0.09
S_{time}	0.21→0.32	Constructor	Construction pace adjustment	–0.14
			Resource scheduling	+0.25
S_{hum}	0.49→0.82	User	Comfort preference	+0.21
		Designer	Green design intensity	+0.12
S_{cul}	0.46→0.74	Designer	Functional zoning adjustment	+0.28
			Green design intensity	+0.11
S_{safe}	0.91→0.95	Constructor	Construction pace adjustment	+0.04
			Resource scheduling	-0.01
S_{env}	0.61→0.83	Designer	Green design intensity	+0.87
		Government	Planning intensity / FAR control	+0.15

4.3. ABLATION STUDY

To systematically evaluate the individual and combined contributions of the proposed framework components, an ablation study was conducted with MAPPO, CFR, and

HRL as the core methodological elements. All experimental settings were kept identical across configurations, and each experiment was independently repeated three times. The results reported correspond to the average performance at the final stable training stage, thereby reducing the influence of stochastic variability. Table 5 presents the final performance of different methods.

Table 5. Ablation study results of MAPPO, CFR, and HRL

State	Initial Value	PPO	PPO+CFR	PPO+HRL	PPO+CFR+HRL
S_{func}	0.48	0.74	0.77	0.78	0.81
S_{form}	0.52	0.70	0.72	0.74	0.77
S_{eco}	0.42	0.71	0.73	0.72	0.74
S_{time}	0.21	0.57	0.61	0.63	0.68
S_{hum}	0.49	0.72	0.76	0.78	0.82
S_{safe}	0.61	0.77	0.80	0.81	0.83
S_{env}	0.46	0.70	0.72	0.73	0.74
S_{cul}	0.91	0.94	0.95	0.95	0.95

The results indicate that this proposed framework achieves the best performance across all dimensions, thereby confirming the synergistic benefits of integrating game-theoretic correction mechanisms with hierarchical decision-making structures.

5. Discussion and Conclusion

To further validate the effectiveness of the proposed framework, an expert evaluation was conducted on the model outputs. Five experts with extensive experience in the design of MTPs were invited to perform a blind assessment of the solutions generated by the model. The evaluation results indicate that the framework exhibits robust optimization and coordination capabilities. Specifically, the incorporation of CFR explicitly institutionalizes behavioural regret as a learning constraint, effectively embedding continuous strategic reflection into each policy update. This mechanism suppresses systemic imbalance induced by short-term individual optimality and promotes the evolution of multi-agent interactions toward a relatively stable equilibrium. In addition, the HRL structure establishes a clear mapping between strategic decision-making and execution-level actions. Through top-down objective guidance and bottom-up feedback correction, HRL enhances coordination across different decision-making levels within the architectural programming process.

This study has several limitations. First, the objective functions and strategy boundaries of stakeholders still rely on abstract assumptions. Second, the environmental modelling is primarily based on idealized conditions and does not yet incorporate uncertainties arising from unexpected events. Future research may address these in several directions. One promising avenue is the integration of large language models to more accurately capture stakeholders' cognitive characteristics and strategy evolution. Another is the integration of the framework with digital platforms such as BIM and VR/AR, which would enhance model interpretability and application value.

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